1.)

```
#include <stdio.h>

int main() {
   int choice,exit_flag=0;
   int a, b;
   unsigned long long factorial;
   while (1) {
      // Step 1: Display menu options
      printf("\nMenu:\n");
      printf("a) Enter 1 to Find the factorial of a number\n");
      printf("b) Enter 2 to Display the remainder on dividing two numbers\n");
      printf("c) Enter 3 to Display the quotient on dividing two numbers\n");
      printf("d) Enter 4 to Find whether the given number is prime or not.\n");
      printf("e) Enter 0 to exit from the menu.\n");
      printf("Enter your choice: ");

      // Step 2: Read the input choice from the user
      scanf("%d", &choice);

      // Step 3: Perform operations based on choice
      // Repeat until user choice is 0
      switch (choice) {
         case 1:
               factorial=1;
               printf("Enter a number: ");
               scanf("%d", &a);
               for (int i=1;i<=a;i++)
                  factorial*=i;
               printf("%llu",factorial);
               break;

         case 2:
               printf("Enter two numbers separated by space: ");
               scanf("%d %d", &a, &b);
               printf("Remainder of %d / %d = %d\n", a, b, a % b);
               break;

         case 3:

               printf("Enter two numbers separated by space: ");
               scanf("%d %d", &a, &b);
               printf("Quotient of %d / %d = %d\n", a, b, a / b);
               break;
```

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        case 4:

                int flag=0;
                printf("Enter a number: ");
                scanf("%d", &a);
                if ((a==0) || (a==1)){
                   printf("non prime");
                   break;}
                else if ((a==2 || a==3)){
                    printf("Prime");
                    break;
                 }
                for(int i=2;i*i<=a;i++)
                {
                    if (a%i==0){
                    printf("non prime");
                    flag=1;
                    break;
                    }

                }
                 if (flag==0)
                 printf("prime");
                 break;
        case 0:
                 exit_flag = 1;
                 printf("Exiting the menu.\n");
                 break;

        default:
                 printf("Invalid choice!\n");
                 break;
      }
      if(exit_flag==1){
      break;
      }

   }

   return 0;
}
```

2.)

```
#include <stdio.h>

int main() {
    int x1, y1, x2, y2, x3, y3;
    int a, b, c, result;

    // Input two points (x1, y1) and (x2, y2)
    printf("Enter point (x1, y1): ");
    scanf("%d %d", &x1, &y1);
    printf("Enter point (x2, y2): ");
    scanf("%d %d", &x2, &y2);

    // Calculate coefficients a, b, and c
    a = y1 - y2;
    b = x2 - x1;
    c = x1*y2 - x2*y1;

    printf("The straight line equation is: %dx + (%d)y + (%d) = 0\n", a, b, c);

    // Read point (x3, y3)
    printf("Enter point (x3, y3) to check its position relative to the line: ");
    scanf("%d %d", &x3, &y3);

    // Check the position of the point
    result = a*x3 + b*y3 + c;
    if (result==0){
        printf("The point (x3, y3) is on the line.\n");
    }
    else if (b*result > 0) {
        printf("The point (x3, y3) is above the line.\n");
    } else if (b*result < 0) {
        printf("The point (x3, y3) is below the line.\n");
    } else {
        printf("The point didn't lie on the line and we can't comment whether the point is above or
below the line as the line is vertical line with infinite slope");
    }

    return 0;
}
```

3.)

8 4 5

4.)

```
#include <stdio.h>

int main() {
    int n, num;
    int largest = -32768;     // take possible min value based on the data type.
    int secondLargest = -32768;  // take possible min value based on the data type.

    printf("Enter the total number of numbers: ");
    scanf("%d", &n);

    printf("Enter %d numbers:\n", n);

    for (int i = 0; i < n; i++) {
        scanf("%d", &num);

        // If current number is greater than largest
        if (num > largest) {
            // Then the current largest becomes second largest
            secondLargest = largest;
            // And the current number becomes largest
            largest = num;
        }
        // If current number is in between largest and second largest
        else if (num > secondLargest && num != largest) {
            // Then the current number becomes second largest
            secondLargest = num;
        }
    }

    if (secondLargest == -32768) {
        printf("There is no second largest number or all numbers are same.\n");
    } else {
        printf("The second largest number is: %d\n", secondLargest);
    }

    return 0;
```

```
}
```

5.)

5,6 and GCD

6.)

```
#include <stdio.h>

int main() {
        int sum=0,val;
        printf("enter the elements of 5*5 matrix");
        for(int i=0;i<5;i++){
            for(int j=0;j<5;j++){
                scanf("%d",&val);
                if (i==j){
                    sum+=val;
                        }
            }
        }
        printf("%d",sum);
        return 0;
}
```

7.)

```
#include <stdio.h>

int main() {
    printf("enter number");
    int num;
    scanf("%d",&num);

    int result = 0, multiplier = 1;
    while (num > 0) {
        if (num % 10 != 0) {
            result += (num % 10) * multiplier;
            multiplier *= 10;
        }
        num /= 10;
    }
    printf("%d",result);
```

```
}
```